

Laporan Minggu 4

Monitoring, Keamanan, dan Optimasi
Final Project Cloud Computing - Kelompok 11
Cloud-Based Data Processing and Monitoring Platform on Microsoft Azure

Nama	Peran
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Tujuan

Minggu 4 berfokus pada operasional sistem: monitoring, alerting, centralized logging, security audit, backup/recovery, dan optimasi biaya.

Monitoring dan Alerting

Area	Implementasi	Tujuan
Application Insights	Terhubung ke Azure Functions	Request log, latency, exception
Azure Monitor	Alert rule operasional	Deteksi error dan performa
Action Group	Email notifikasi tim	Respons incident
Dashboard aplikasi	Statistik, tabel, dan chart	Monitoring data hasil proses

Alert Rule Yang Disiapkan

Alert	Target	Threshold
Function 5xx Error	Azure Function App	HTTP 5xx > 5 dalam 5 menit
Function Latency	Azure Function App	Average response time > 2 detik
VM CPU High	VM Web	CPU rata-rata > 80 persen selama 5 menit

Security Baseline

Area	Kontrol	Status
Frontend secret	Browser hanya memanggil /api	Selesai
Function key	Cloudflare env server-side	Selesai
Cosmos secret	Azure Key Vault	Selesai
Password	PBKDF2 hash	Selesai
Role	Register publik role user	Selesai
Data isolation	Filter owner_user_id	Selesai
Repo hygiene	.env/.tfvars/state/key di-ignore	Selesai
VM SSH	Batasi IP admin	Perlu hardening

Backup dan Recovery

- Cosmos DB mengandalkan backup platform Azure sesuai konfigurasi account.
- Blob Storage raw-data dapat digunakan untuk reprocessing.
- Source code tersimpan di GitHub sebagai version control.
- Simulasi recovery disarankan: upload file kecil, verifikasi Cosmos DB, export sample record, lalu proses ulang file.

Cost Optimization

Optimasi	Dampak
Cloudflare Pages	Mengurangi kebutuhan VM untuk frontend
Azure Functions Consumption	Mengurangi biaya backend idle
Cosmos DB Serverless	Cocok untuk traffic kecil proyek

Optimasi	Dampak
Storage lifecycle policy	Mengontrol pertumbuhan file mentah
Matikan VM saat tidak demo	Mengurangi biaya compute

Screenshot Yang Perlu Diambil Dari Portal

Bukti	Lokasi
Application Insights overview	Azure Portal > Application Insights
Alert rules dan action group	Azure Monitor > Alerts
Cost breakdown	Azure Cost Management
Security posture	Defender for Cloud
Backup/recovery evidence	Cosmos DB / Storage

Catatan

Sebelum screenshot portal dimasukkan ke laporan, pastikan tidak ada function key, Cloudflare API token, connection string, access key, password, file tfvars, atau tenant/subscription ID penuh yang terlihat.

Bukti Azure - Application Insights dan Metrics

Week 4 Evidence - Monitoring

Azure Application Insights and metrics snapshot

Generated 2026-05-17 14:23 UTC

Application Insights

Name	func-backend-monitoring-k11
Location	southeastasia
App type	web
Provisioning	Succeeded

Function App

Name	func-backend-monitoring-k11
State	Running
Kind	functionapp,linux
Host	func-backend-monitoring-k11.azurewebsites.net

Latest Metrics

Requests total	0.000
HTTP 5xx total	0.000
Avg response time	No recent data seconds
VM avg CPU	0.332 percent

Source: Azure CLI sanitized output. No function key, token, access key, or connection string shown.

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Halaman 4

Bukti Azure - Alert Rules dan Action Group

Week 4 Evidence - Alerting

Azure Monitor action group and metric alert rules

Generated 2026-05-17 14:23 UTC

Action Groups

Application Insights Smart Detection enabled=True shortName=SmartDetect
ag-kelompok11-ops enabled=True shortName=k11ops

Metric Alerts

Alert name	Enabled	Severity	Window	Evaluation	Description
alert-function-5xx-errors	True	2	PT5M	PT1M	Alert jika Azure Function menghasilkan HTTP 5xx berulang
alert-function-latency	True	3	PT5M	PT1M	Alert jika rata-rata response time Azure Function lebih
alert-vm-cpu-high	True	3	PT5M	PT1M	Alert jika CPU VM rata-rata lebih dari 80 persen.

Source: Azure CLI sanitized output. Notification email is intentionally not displayed.

Bukti Azure - Cosmos DB dan Blob Storage

Week 4 Evidence - Backup and Data

Cosmos DB backup policy and Blob Storage containers

Generated 2026-05-17 14:23 UTC

Cosmos DB

Name	cosmos-kelompok11-monitoring
Location	Southeast Asia
Consistency	Session
Backup policy	Periodic
Provisioning	Succeeded

Cosmos Containers

users	/email
telemetry-data	/deviceId

Storage Account Containers - stfuncmonitoringk11

azure-webjobs-hosts	publicAccess=private
azure-webjobs-secrets	publicAccess=private
function-releases	publicAccess=private
github-actions-deploy	publicAccess=private
raw-data	publicAccess=private
scm-releases	publicAccess=private

Source: Azure CLI sanitized output. Access keys and connection strings are not displayed.

Bukti Azure - Security dan Cost Management

Week 4 Evidence - Security and Cost

Defender for Cloud pricing status and Cost Management usage

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Defender for Cloud / Security Pricing

VirtualMachines	Free
AppServices	Free
StorageAccounts	Free
KeyVaults	Free
CosmosDbs	Free
CloudPosture	Free
FoundationalCspm	Standard

Cost Management Usage - RG-Kelompok11

Microsoft.Storage	150 usage rows
Microsoft.Compute	30 usage rows
microsoft.web	23 usage rows
microsoft.documentdb	19 usage rows
Microsoft.Network	18 usage rows
microsoft.network	17 usage rows
microsoft.compute	11 usage rows
Microsoft.KeyVault	6 usage rows

Note: Azure for Students/Consumption API returned cost fields as None in CLI.

Source: Azure CLI sanitized output. Subscription IDs, tenant IDs, and billing identifiers are omitted.